

The AI Infrastructure Imperative

A Strategic Advisory for Data Center Capital Deployment

A data center CEO came to us with one question:

“Where do I spend my money, and when?”

This presentation is our answer.

Vince · Johanes · Yabes · Ireen · Blessing

ClearVision Ventures · Final Research Presentation · April 2026

CONTEXT

Five Forces. One Decision.

The data center company's future is shaped by five forces converging at the same moment. Getting any one of them wrong compounds into years and billions lost.

01

Power Infrastructure

GaN and HTS

Vince

Lead Analyst

02

Software and Orchestration

DCIM, MLOps, Security

Johanes

Lead Analyst

03

Silicon and Accelerators

GPU, ASIC, Photonics

Yabes

Lead Analyst

04

Photonic Interconnects

The copper to light transition

Blessing

Lead Analyst

05

Cooling and Thermal Systems

Waste heat and chip cooling

Ireen

Lead Analyst

The Core Thesis

\$650B is committed. 49GW is missing. Half the builds are delayed. The constraint is not demand, it is infrastructure readiness. Every dollar deployed in the wrong layer at the wrong time is a dollar that does not compound. Our five research tracks converge on one answer: what to build, what to buy, and when to act.

THE FORCING FUNCTION

The Power Crisis Is Physics, Not Policy

AI rack density has crossed a threshold where copper infrastructure is not just inefficient. It is mathematically incompatible with what is being asked of it.

300kW+

AI rack power today
vs ~10kW legacy

1 MW

Per rack by 2029
Vertiv projection

49 GW

US power shortfall
by 2028 Morgan Stanley

50%

Planned 2026 builds
already delayed or canceled

The I²R Wall

Power loss scales with current squared. At 12V and 300kW, resistive heat alone forces multiple redundant cable runs and expands cooling loads before a single computation starts. Raising voltage to 800V cuts I²R losses by over 99%.

Copper Is Running Out

Wood Mackenzie: 304,000 tonne deficit in 2025, widening in 2026. UBS: 400,000 tonne shortfall. S&P Global calls the projected 10 million metric ton gap by 2040 a systemic risk. A single 100MW data center absorbs 27 to 33 tonnes of copper before upstream grid work begins.

The Grid Cannot Keep Up

New transmission lines take 10 or more years to permit. New substations take 3 to 7 years. Morgan Stanley projects a 126GW increase in global data center power demand through 2028. Gartner: 40% of AI data centers constrained by power availability by 2027.

Bloomberg and Sightline Climate
April 2026

~50%

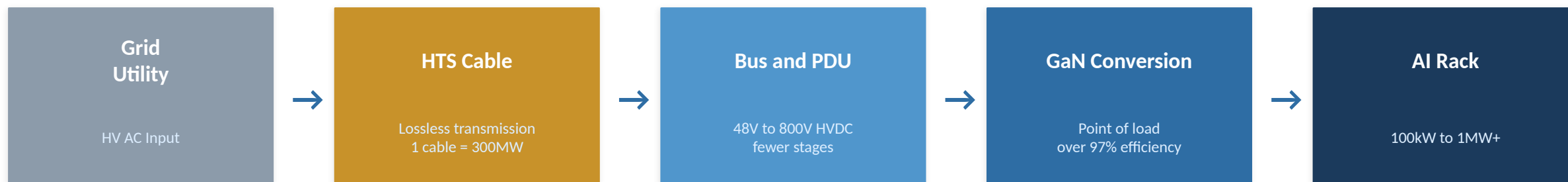
of planned 2026 US data center builds are
already delayed or canceled

Root cause: electrical components with 3 to 7 year
lead times. Money is not the constraint. Power
infrastructure is.

**GaN and HTS
unblock all of it.**

THE SOLUTION STACK

GaN Today. HTS by 2027. One Complementary Power Stack.



GaN — Deploy Now

- 98.5% peak efficiency on Navitas 10kW platform at 1MHz switching (February 2026)
- NVIDIA Kyber 800V HVDC architecture locking in across hyperscaler fleets
- 400M ICs shipped with zero field failures confirms hyperscale procurement confidence
- Infineon 300mm wafer production delivers cost parity with silicon
- IISc 2025 gate stack cuts leakage 10,000 times and retires qualification risk
- ISO 14040 certified: saves over 10 million tonnes of CO₂ per year at data center scale

HTS — Pilot 2027, Scale by 2030

- 1.7% residual loss versus 3 to 5% for copper confirmed at 100MW scale (2022 study)
- VEIR 3MW OCP demo November 2025 and world's first HTS powered rack prototype
- Microsoft invested \$75M into VEIR and confirmed factory testing of the 3MW cable
- MetOx Project Arch breaks ground 2025 with \$80M DOE backing, first US HTS wire plant
- University of Houston PNAS March 2026 sets 151K ambient pressure superconductivity record
- November 27 2026: China gallium ban suspension expires. Market is not pricing this risk.

THE SOFTWARE STACK

AI Broke the Old Tools. Every Layer Is Being Rebuilt.

Legacy DCIM, MLOps platforms, and security tooling were designed for predictable CPU workloads. AI data centers are not that, and the software market is resetting accordingly.

Layer	Maturity	OS Risk	Hyperscaler Threat	VC Rating	Recommendation	Why It Matters
L1 — Infrastructure	High	High	Medium	2 of 5	Avoid	Kubernetes won. VMware disruption benefits open source, not venture startups.
L2 — DCIM and Energy	Medium	Low	Low	4 of 5	Invest	No category leader. Energy mandates create non-discretionary demand. Highest ROI for operators.
L3 — AI and ML Workloads	Low	Medium	Low	5 of 5	Top Priority	MLOps grows from \$2.33B to \$25.9B by 2034. 2026 is the inflection from training to inference.
L4 — Security	Medium	Low	Medium	4 of 5	Invest	EU AI Act enforcement started 2025. No vendor has a purpose built AI compliance module yet.
L5 — Observability	High	Medium	High	2 of 5	Selective	Datadog and Cisco Splunk own the space. AI model observability is the only remaining whitespace.

\$72.3B

SDDC market 2024 growing at 30.4% CAGR through 2033

28.9%

MLOps CAGR through 2034, fastest defined software category

\$81B

IT operations management spending by 2028 per Gartner

20 to 30%

IT support cost reduction from AIOps deployment per BCG

INVESTMENT CONVICTION

Three Software Categories Worth Owning Right Now

The highest returns will come from software where AI has made the existing tools no longer good enough and no clear replacement vendor has emerged yet.

01

AI Native DCIM and Energy Optimization

Seed to Series B

No category leader has emerged. Top 5 DCIM players hold only 26% combined market share.

DCIM market at \$3.7 to \$4.7B growing at 15 to 23% CAGR. Substantial whitespace ahead.

Energy regulation through EU EED and EPA creates non-discretionary procurement demand.

Google DeepMind AI cooling achieved 40% reduction in cooling energy. Thesis validated.

One percent efficiency gain equals \$200M annual savings across North America and Europe.

Hyperscaler threat is low. They have no incentive to build software for third party operators.

02

Inference Optimization and Multi Cloud MLOps

Series A to B

MLOps market grows from \$2.33B in 2025 to \$25.9B by 2034 at 28.9% CAGR.

2026 is the structural inflection point from model training to inference deployment at scale.

Reasoning models use 5 to 10 times more compute per query. Inference cost is urgent.

Lambda raised \$1.5B Series E in Q4 2025 confirming deep investor conviction in the category.

Multi cloud MLOps is structurally protected from hyperscaler substitution.

No clear winner has emerged in inference optimization. The market is still open.

03

Zero Trust and AI Compliance Automation

Seed to Series A

Data center security market at \$17.9B in 2025 growing at 11.9% CAGR.

EU AI Act full enforcement began 2025. Compliance spend is now mandatory, not optional.

No vendor has purpose built AI compliance modules. The sub segment has no published TAM.

Cyber insurers now require AI audit capabilities, creating a second independent demand driver.

Very strong net revenue retention profile once embedded in a customer's operations.

Regulation driven lock in from AI Act and NIS2 creates durable, defensible revenue.

THE SILICON LAYER

The GPU Is Not the Story. The Interconnect Is.

Silicon accelerators hold 60 to 70% of AI server bill of materials. But the real ceiling is no longer transistor count. It is bandwidth, thermals, and power delivery. Whoever relieves those constraints captures the value.

\$650B

Global DC capex by 2030, 2.5 times
2023

35%+

CAGR of AI accelerator spend
through 2028

60 to 70%

Share of AI server BOM held by
silicon

2x

Power density per rack versus
2020, the real ceiling

Moore's Law Did Not Die. It Moved.

Transistor count is no longer the constraint. Power, interconnect bandwidth, and thermal density now define competitive position. The data center companies that understand this distinction will spend their capex correctly. The ones that do not will build infrastructure that is obsolete before it opens.

Hyperscalers Are Spending Correctly

Over \$200B of 2025 hyperscaler capex flows directly into silicon and silicon adjacent infrastructure. Google TPU, Amazon Trainium, Microsoft Maia, and Meta MTIA are not chip programs. They are power and interconnect optimization programs wearing silicon as a costume.

The Binding Constraint Is Bandwidth

Electrical interconnects at 224 Gbps cannot transmit data more than one meter without active signal regeneration. This is the skin effect and it is fundamental physics. At 18,000 GPU cluster scale, the retimer chips required to regenerate copper signals consume 480kW solely to move data between chips.

The Investment Angle

- Lead a \$40M Series B into a top tier silicon photonics startup.
- Entry at approximately \$180M pre money valuation.
- TAM of \$15 to \$20B by 2030 at roughly 30% CAGR.
- Target 6 to 8 times MOIC over 5 to 7 years.
- Exit through strategic M&A from NVIDIA, Broadcom, Intel, or a late stage growth round.
- Every hyperscaler roadmap from 2026 onward assumes optical interconnects. Demand is pulled, not pushed.

ClearVision can own the bandwidth layer while the rest of the market is still chasing compute.

INVESTMENT THESIS

Lead the \$40M Silicon Photonics Round Before the Market Consensus Arrives

Optical interconnects are the picks and shovels of the AI decade. Every GPU cluster from 2026 forward is designed around them. The physics moat is real and the exit market is already validated.

Entry

\$40M Series B at ~\$180M pre money. Secure board seat and pro rata rights through Series C and D.

Market

TAM of \$15 to \$20B by 2030 at 30% CAGR. \$36B combined scale up interconnect market by 2030 per 650 Group.

Return

6 to 8 times MOIC target over 5 to 7 years. Strategic M&A from NVIDIA, Broadcom, or Intel as primary exit.

30 / 60 / 90 Day Action Plan

30 Days

Shortlist three leading silicon photonics candidates from the Ayar Labs, Lightmatter, and POET tier. Begin technical diligence on optical chip architecture and packaging yield data.

60 Days

Complete IP audit and confirm cross ecosystem compatibility with both NVLink Fusion and UALink 2.0. Require named hyperscaler design wins, not roadmap presentations.

90 Days

Issue term sheet and secure hyperscaler letter of intent. Confirm foundry booking at TSMC and second source packaging agreement before closing.

Key Risks and Mitigation

Long commercial cycles and hyperscaler insourcing risk are addressed through milestone tranching, strategic LP co invest, and requiring second source IP licensing as a condition of investment.

THE PHYSICS OF COPPER FAILURE

Copper Has Not Approached Its Limits. It Has Exceeded Them.

At 224 gigabits per second per lane, the speed required for modern AI interconnect, copper cannot transmit data more than one meter without active regeneration. This is not an engineering problem awaiting a clever solution. It is the skin effect: a fundamental property of electromagnetism.

480kW

Wasted in an 18,000 GPU cluster
just to move data between chips

55pJ

Energy per bit on copper vs under
5pJ on co packaged optics

\$175M

Annual electricity savings at
hyperscale from photonic adoption

7 months

Payback period on photonic
premium from operating savings
alone

The Energy Case Is Positive on Operating Cost Alone

Copper at AI speeds consumes 55 picojoules per bit. Co packaged optics consume fewer than 5. At one million active data connections the annual electricity savings exceed \$175M. The capital premium for photonic interconnect over copper is approximately 1.9% of total cluster cost. It pays for itself in seven months.

The Supply Chain Constraint Creates Pricing Power

Global demand for indium phosphide devices reached 2 million units in 2025 while production capacity was only 600,000. A 70% supply demand gap. NVIDIA has pre allocated EML laser capacity at Lumentum and Coherent through 2027. Companies without a supply agreement face delays regardless of budget.

The Exit Market Has Already Been Validated

Marvell acquired Celestial AI for \$3.25B in February 2026. The acquisition occurred because hyperscalers were already designing with the technology. The next Celestial AI is being built now and the companies building it are still accessible at venture valuations.

UALink 2.0 Opens a Second Market

The UALink Consortium, backed by AMD, Intel, Google, Microsoft, Meta, Amazon, and Broadcom, released version 2.0 on April 7 2026. This creates a distinct photonics market that did not exist before 2026. Photonic startups building interconnect agnostic optical engines gain access to this market without requiring customers to leave NVIDIA.

Deployment Timeline

2026

CPO in initial production at Broadcom and NVIDIA. UALink 2.0 released. InP shortage most acute.

2027

UALink silicon in commercial production. CPO scaling to 30% of cluster interconnect. Laser supply beginning to ease.

2028

Optical circuit switching entering production for building scale fabric. First fully optical AI training clusters.

2030+

CPO as default interconnect for all new AI cluster builds. Energy per AI token down 70% from 2025 baseline.

WHERE TO INVEST IN PHOTONICS

Three Layers with the Highest Conviction

The transition from copper to light is not a thesis requiring further validation. It is a deployment requiring capital. The three layers below are where the value concentrates.

Layer 2

Photonic Chip Design

Venture Fit: Very High

Companies

Lightmatter, Ayar Labs, POET Technologies

Why

Rarest engineering capability in the stack. Proprietary architectures with hyperscaler design wins command the highest acquisition premiums. The \$3.25B Celestial AI exit is the benchmark.

How to Enter

Require cross ecosystem compatibility with both NVLink Fusion and UALink. Companies locked to a single ecosystem address at most 70% of the available market.

Layer 4

Advanced Packaging and Integration

Venture Fit: Very High

Companies

Ayar Labs, Lightmatter, specialist packaging houses

Why

The hardest engineering problem in the full stack. Very few companies can do this at production yield. High switching cost once embedded in a customer supply chain makes churn nearly impossible.

How to Enter

Require yield data from existing production runs. The gap between a working prototype and 90% yield production is where most photonic companies fail commercially.

Layer 1

Laser Materials Innovation

Venture Fit: High

Companies

POET Technologies, emerging III-V on silicon startups

Why

Deepest structural constraint in the full stack. Any startup solving the indium phosphide shortage addresses a bottleneck affecting every photonic product in the entire market simultaneously.

How to Enter

Secure InP laser allocation with Coherent or Lumentum for 2027 to 2028 delivery on behalf of portfolio companies as a condition of investment. Budget alone will not secure supply.

What to Avoid

Do not invest in copper infrastructure or DSP based transceiver manufacturers. Do not accept roadmap slides without foundry booking confirmation and production yield data. Do not commit to a single CPO vendor

THE THERMAL LAYER

Heat Is the Hidden Tax on Every Watt You Generate

The AI era data center has transitioned from a demand driven system to a constraint driven one. Power, cooling, water usage, and permitting now define the boundaries of feasible expansion. Thermal management is no longer a cost center. It is a value creator.

20 to 25%

Of total facility power consumed by traditional air cooling systems

40%

Cooling energy reduction from AI driven thermal optimization per Google DeepMind

15 to 21%

Greenhouse gas reduction from liquid cooling plates per Microsoft Nature 2025

52%

Water usage reduction with two phase immersion cooling per Microsoft 2025

From Facility Level to Chip Level

Traditional air cooling manages heat at the room and rack level. At 300kW per rack, heat must now be addressed at the chip package level. Corintis Series B at approximately \$53M is validated by hyperscale partnerships. This represents the directional shift from CRAC units to direct liquid to chip paths.

Regulation Is Creating Non Discretionary Demand

European Energy Efficiency Directive mandates are transforming waste heat from an unmanaged byproduct into a required output. This is not a voluntary sustainability initiative. It is a compliance requirement with enforcement timelines. Companies that build waste heat recovery into new facilities now avoid costly retrofits later.

Thermal Architecture Determines Power Efficiency

A data center's PUE is largely determined by its thermal architecture. Google's best in class PUE of 1.09 versus an industry average of 1.58 represents roughly 45% more usable compute per watt. At 945 TWh of projected global data center electricity by 2030, that gap is worth tens of billions annually.

Modular Infrastructure Removes Deployment Bottlenecks

Prefabricated power and cooling systems allow for faster capacity scaling and reduced construction complexity. Flexnode Series A in 2026 exemplifies this approach, offering integrated modular data halls with significant reductions in deployment lead time versus traditional construction.

Why It Matters Now

- AI racks exceeding 100kW make traditional air cooling physically incompatible, not just inefficient.
- EU EED mandates on waste heat recovery are enforcement stage. Not optional.
- Thermal management solutions compound with power and silicon efficiency gains across the full stack.
- PUE of 1.09 versus 1.58 industry average equals 45% more compute per watt. That is a structural cost advantage.
- Companies solving thermal at chip level will capture operating cost savings across every generation of future hardware.

INVESTMENT CONVICTION

Four Thermal Investment Lanes Ranked by Urgency

Thermal is the category where regulatory enforcement, technical necessity, and economic value creation converge at the same time. The lanes below are ordered by how soon capital should move.

01

Waste Heat Recovery and Thermal Reuse

Act Now

2026 to 2028

European Energy Efficiency Directive creates mandatory non discretionary demand. Compliance required, not optional. Karman Industries: Series A at \$20M in 2025 developing modular CO₂ based systems for high density data centers. Opinch: Series B at approximately \$21M offering chemical heat transformation as a complementary technology. Category offers moderate capital requirements and clear acquisition pathways through energy infrastructure players. Companies investing now avoid costly retrofits when regulation reaches enforcement in their jurisdiction.

02

Advanced Thermal Materials and Chip Level Cooling

Invest Now

2026 to 2030

As rack densities exceed 100kW, cooling must move from room level to chip package level. No alternative path. Corintus: Series B at approximately \$53M validated by hyperscale partnerships. Represents the leading approach. ZutaCore and GRC offer alternative approaches illustrating diversity of technical paths in this category. Capital inflows are accelerating. Many leading companies are advancing quickly beyond early stage windows. Timing and valuation discipline are critical. The entry window for best in class companies is narrowing.

03

Modular Prefabricated Power and Cooling Infrastructure

Selective Entry

2026 to 2028

Prefabricated systems address deployment speed, the most acute operational constraint in data center expansion. Flexnode: Series A in 2026, offering integrated modular data halls with dramatically reduced deployment timelines. Strategic investment by Eaton introduces limitations in exit flexibility requiring careful investor evaluation. Best suited to investors with infrastructure relationships and operational co investment partners.

04

Thermal Energy Storage

Monitor for 2028

2028 to 2032

Ice based cooling systems enable load shifting and improve grid interaction during peak demand periods. Nostromo Energy provides the reference implementation model for this category. Limited early stage data center specific companies currently available limits immediate venture opportunity. Position this as a longer term monitoring priority rather than an immediate deployment decision.

SYNTHESIS

Five Layers. One System. You Cannot Optimize One Without the Others.

A data center CEO cannot solve power without understanding its thermal consequences. Cannot solve cooling without the right software to measure it. Cannot benefit from silicon advances without the interconnects to move data at that speed. These five tracks are one compounding system.



Power and Cooling

Every watt delivered by GaN at 97% efficiency still generates 3% heat at scale. At 1MW per rack that is 30kW of thermal load per rack before cooling is considered. Power architecture decisions lock in cooling requirements before construction begins.

Cooling and Software

AI native DCIM is the feedback loop that connects thermal sensors to workload scheduling. Without software that understands thermal state in real time, even the best cooling infrastructure operates at static set points and leaves efficiency unrealized.

Photonics and Silicon

Silicon photonics is not a separate category from the accelerator layer. It is the interconnect that makes GPU clusters function at full capacity. A data center investing in NVIDIA Vera Rubin NVL576 without securing photonic interconnect supply has built a runway with no landing lights.

The data center company that wins the next decade is not the one with the most capital. It is the one that sequences these five investments correctly and understands their dependencies before committing. That sequencing is what the next two slides make explicit.

KNOW WHAT CAN GO WRONG

Six Risks That Require Active Management

Strong theses fail when risks are identified too late to act on them. Each of the six risks below has a known trigger, a knowable timeline, and a specific mitigation available right now.

Risk	Probability	Impact	Who It Hits	When	Mitigation
China Gallium Ban Reinstated	High	Severe	All GaN dependent builds	November 27 2026	Invest in MTM Critical Metals for domestic gallium recovery. Navitas and GlobalFoundries US partnership reduces but does not eliminate exposure.
HTS Commercialization Delayed Beyond 2028	Medium	High	Hyperscale campus builds	2028 onward	GaN bridges the gap at the conversion layer. Maintain copper contingency for campus feed until pilot data confirms VEIR commercial readiness.
InP Laser Shortage Extends to 2028	High	High	All photonics dependent builds	Immediate	Secure multi year supply agreements with Coherent or Lumentum now. Budget alone will not secure allocation. POET Technologies as structural hedge.
Hyperscaler Full Vertical Integration	Medium	Medium	Merchant component vendors	2027 to 2029	Invest only in companies with cross ecosystem compatibility across both NVLink and UALink. Monitor patent filings as leading indicator of insourcing intent.
AI Efficiency Gains Reduce Near Term Power Demand	Low	Low to Medium	GaN and HTS investment thesis	Rolling	Power infrastructure demand is structural and tied to data center construction, not model efficiency. Historical pattern: efficiency gains expand AI use, they do not reduce it.
Cooling Regulation Tightens Faster Than Expected	Medium	High	Legacy data center operators	2026 to 2027	Treat EU EED compliance as the minimum bar, not the ceiling. Build thermal audit and retrofit pipeline into capital planning for all existing facilities.

WHEN TO SPEND WHAT

The Capital Deployment Roadmap

The architecture choices made in the next 12 months will lock in economics for 15 to 20 years. This roadmap sequences the five investment areas by urgency, dependency, and return horizon.

Now through 2026

Power

Specify GaN PSUs into all new builds immediately. NVIDIA Kyber 800V is the standard. Miss this window and you are retrofitting in 2028.

Software

Deploy AI native DCIM software across all facilities. No category leader means the best price and the best terms are available right now.

Photonics

Secure indium phosphide laser allocation with Coherent or Lumentum for 2027 to 2028 delivery. Budget alone will not get supply.

Supply Chain

Conduct gallium supply chain audit across all GaN vendors before November 27 2026 ban suspension expiry.

2026 through 2027

Thermal

Pilot and fund waste heat recovery systems in European facilities where EU EED mandate enforcement is nearest.

Photonics

Source silicon photonics Series B candidates. UALink 2.0 opens a second market in late 2026. Entry window is now.

HTS

Evaluate VEIR commercial pilot. 2027 commercial launch confirmed. First mover advantage on data center HTS delivery.

Software

Begin procurement and integration of inference optimization tooling. 2026 is the training to inference inflection point.

2027 through 2030

HTS

Scale HTS campus power delivery following pilot data. MetOx wire supply and VEIR system deployment are complementary.

Thermal

Layer in chip level thermal materials investment as Corintis tier companies approach Series C and beyond.

Software

Deploy full AI compliance automation suite as EU AI Act enforcement matures and cyber insurer requirements escalate.

Silicon

Evaluate on chip photonics chipllets as they approach production readiness. Do not commit capital before 2028 yield evidence.

Our Recommendation to ClearVision Ventures

What to Build, When to Spend, and Where the Value Lives

WHAT

Build or operate a next generation AI data center that is competitive in 2027 to 2030, not 2024. The infrastructure decisions made today are a 15 to 20 year commitment.

WHEN

The window is 2026. Not 2027. The architecture choices made in the next 12 months lock in your economics for two hardware generations. Waiting costs compounding margin.

WHY

\$650B is committed but blocked. 49GW is missing. Copper is running out. Software tools are obsolete. Interconnects are failing at current speeds. The CEO who solves these five constraints in sequence will operate at half the cost of the CEO who waits.

WHICH

Layer	Act Now	Watch and Prepare	Wait for Evidence
Power	GaN PSUs in all new builds immediately	HTS campus pilots through VEIR	Full HTS buildout until 2028 pilot data confirms
Software	AI native DCIM across all facilities	Inference optimization tooling now	Full AI compliance suite until regulation matures
Photonics	Secure InP laser supply allocation	Lead silicon photonics Series B	On chip chiplets until 2028 yield evidence
Cooling	Waste heat recovery in EU facilities	Chip level thermal materials investments	Thermal storage until 2028 to 2030

Vince

GaN is not optional, spec it in now. HTS is the 2027 bet that unlocks megawatt scale campus delivery.

Johanes

The software stack needs a rebuild. AI native DCIM is the highest ROI software investment available today.

Yabes

The interconnect layer is where the alpha hides. Lead the silicon photonics round before the market catches up.

Blessing

Copper is over. The \$3.25B Celestial AI exit proved the thesis. The next one is being built right now.

Ireen

Thermal is not a cost center, it is a value creator. Waste heat recovery pays back in 12 to 18 months.

Thank You

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Vince · Johaness · Yabes · Ireem · Blessing

*“The constraint is not capital.
It is knowing where to deploy it.”*

Vince

Power and Infrastructure

Johaness

Software and Orchestration

Yabes

Silicon and Accelerators

Ireem

Cooling and Thermal

Blessing

Photonic Interconnects